

Note on statistics:

Here we focus on the impacts of smoke on $\text{PM}_{2.5}$ and O_3 concentrations

$\text{PM}_{2.5}$ due to smoke:

To identify the presence of smoke in a region, we use the NOAA Hazard Mapping System (HMS) fire and smoke product (Ruminski et al., 2011; Kaulfus et al., 2017; Buysse et al., 2019). It is based on multiple geostationary satellites and gives a smoke density as high, medium or low. However, as the HMS product is derived from satellite data, it does not provide any vertical information on smoke plumes. Our previous analysis (Buysse et al., 2019) indicates that enhanced $\text{PM}_{2.5}$ at the surface is found on 30-80% of days with an overhead smoke plume, depending on the location. Nevertheless, the HMS smoke product is critical to identify days when regional $\text{PM}_{2.5}$ concentrations are possibly influenced by smoke (Kaulfus et al., 2017). We define any location that is within the boundaries of an HMS smoke plume as $\text{HMS}=1$ and a site with no overhead smoke as $\text{HMS}=0$.

We combine the HMS product with surface $\text{PM}_{2.5}$ to identify the presence of smoke at the surface. But this leaves the question of what $\text{PM}_{2.5}$ value or criteria should we use to define smoke influence? Figure 1 shows the distribution of $\text{PM}_{2.5}$ on $\text{HMS}=0$ and $\text{HMS}=1$ days.

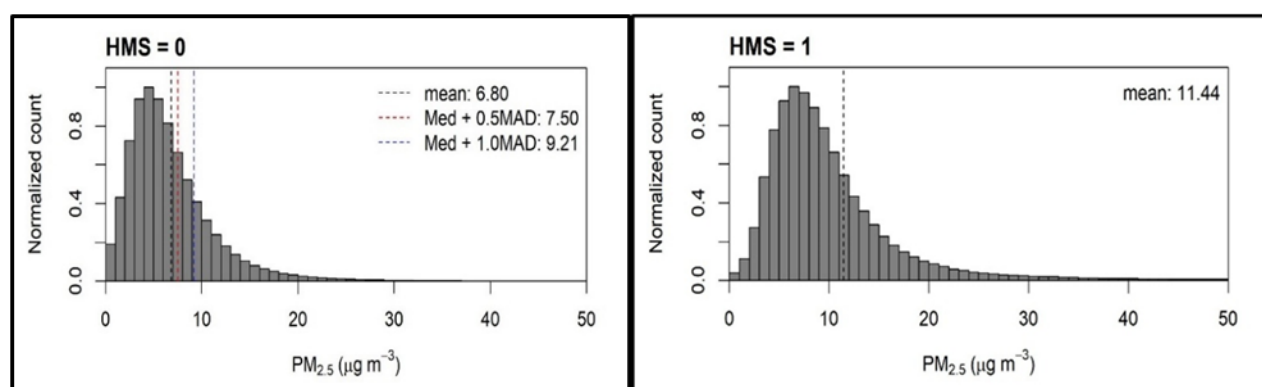


Figure 1. Distributions of daily $\text{PM}_{2.5}$ concentrations for $\text{HMS}=0$ (left) and $\text{HMS}=1$ (right) days. For the $\text{HMS}=0$ distribution the median + 1 and 0.5 median absolute deviations (MAD) are also shown. This plot shows the full year data for all $\text{PM}_{2.5}$ sites. If we use only the April–October data instead, the means are similar (6.6 and 10.9 $\mu\text{g m}^{-3}$, respectively).

While the means of these two distributions are significantly different and the $\text{HMS}=1$ data show a

greater occurrence of high PM_{2.5} days, the distributions have substantial overlap. Our goal is to choose a PM_{2.5}-criteria that is **most likely** associated with smoke and is **unlikely** to be due to local pollution alone. We note that this is different from “outlier” detection in statistics. Since our goal is to identify **any** significant contribution to the PM_{2.5} concentrations, down to a few $\mu\text{g m}^{-3}$, using an outlier method would mean that most moderate smoke days would not be classified as having a smoke influence.

Previously, we used the mean plus one standard deviation (Mean + 1 SD) of a sites’ HMS=0 PM_{2.5} concentrations as a criteria to define smoke influence for the GAM-v1 dataset (Lee and Jaffe, 2024a; 2024b). But the figure above shows that the PM_{2.5} distributions are not normally distributed and so the standard deviation is not a good metric to use for these distributions. Instead, we use a method that is more suited for non-normal distributions called the median absolute deviation (MAD, Leys et al., 2013). The MAD is defined as:

$$\text{MAD} = M_2 |X(i) - M_1|$$

Where $X(i)$ is the distribution of PM_{2.5} observations, M_1 is the median of the PM_{2.5} data and M_2 is the median of the absolute deviations from M_1 . Since PM_{2.5} data have a strong seasonality (highest in winter in the absence of wildfire smoke), the MAD values are calculated for each site and each month separately. This is why it’s necessary to have multiple years of data to identify enough HMS=0 days. Typically, we use a minimum of 4-6 years of data.

We define the PM_{2.5}-criteria as the Median + 0.5 MAD and Median + 1 MAD, which is approximately the 67th and 80th percentiles of the HMS=0 distribution at each site. Note that for the three datasets: GAM-v2 (Lee and Jaffe, 2025) and “Smoke PM_{2.5}” the app provides both PM_{2.5}-criteria (Median + 0.5 MAD and Median + 1.0 MAD). The average PM_{2.5}-criteria (Median + 1 MAD), averaged for all months range from 2.6 $\mu\text{g m}^{-3}$, at very clean sites along the west coast, to 17.9 $\mu\text{g m}^{-3}$ at more polluted urban locations. So, this points out the greater challenge of identifying moderate smoke levels at more polluted urban locations. At the same time, we note that the mean and median PM_{2.5}-criteria are 8.51 and 8.46 $\mu\text{g m}^{-3}$, respectively, so this is not an issue at the vast majority of sites and more than 93% of the 803 O₃ sites (i.e., GAM-v2) we consider have PM_{2.5}-criteria less than 12 $\mu\text{g m}^{-3}$.

Finally, one variable to pay attention to in our datasets is the “Quant_PM2.5”. This gives

the percentile of each PM_{2.5} value, in comparison to the HMS=0 data. The greater is this value, the greater is the confidence that smoke has influenced the PM_{2.5} data. During smoke events, we can see that many adjacent sites will have high values for Quant_PM2.5.

O₃ due to smoke:

The contribution of smoke to surface O₃ is complex. On average, smoke increases surface O₃ concentrations, but there is a large degree of variability due to varying emissions, meteorology during transport, smoke impacts on chemistry and photolysis and the local photochemical environment at the receptor site. Smoke can contribute to urban O₃ concentrations either directly (e.g. O₃ in the smoke plume mixes into an urban area) or in-directly, by enhancing photochemical production. For the in-direct enhancement, it is possible that VOCs in smoke plumes combine with existing urban NO_x to enhance O₃ production. It is also possible that PAN (peroxyacetyl nitrate) present in the smoke plume decomposes at the surface (due to warmer temperatures) and this will also contribute to O₃ production. For all of these reasons, the contribution of smoke to surface O₃ is highly variable and challenging to estimate (Jaffe et al., 2020).

To estimate the smoke contribution to surface O₃ we use a statistical/machine learning approach to predict O₃ **in the absence of smoke**. This is done using “Generalized Additive Models” or GAMs. For each site, the GAMs estimate the MDA8 concentration in the absence of smoke using a variety of predictors. Importantly, the GAMs can include linear, non-linear and categorical relationships, so this means we can include typical met variables (such as daily max temperature, RH, etc.), plus variables such as the trajectory wind vector and speed, day of week, etc. as predictors for the MDA8. The GAMs must be calculated independently at each location since the relationship with the meteorological variables is different at each site. A variety of quality control measures are evaluated, as described in our various papers and publications.

Figure 2, below, shows performance of the GAM-v2 for 803 sites nationally for data from April–October 2019–2024. This plot includes all observed and predicted MDA8 values and has a mean R² of 0.83. The mean R² for the individual GAMs at these 803 sites is 0.75. We define the model residuals as observed MDA8 – GAM predicted MDA8.

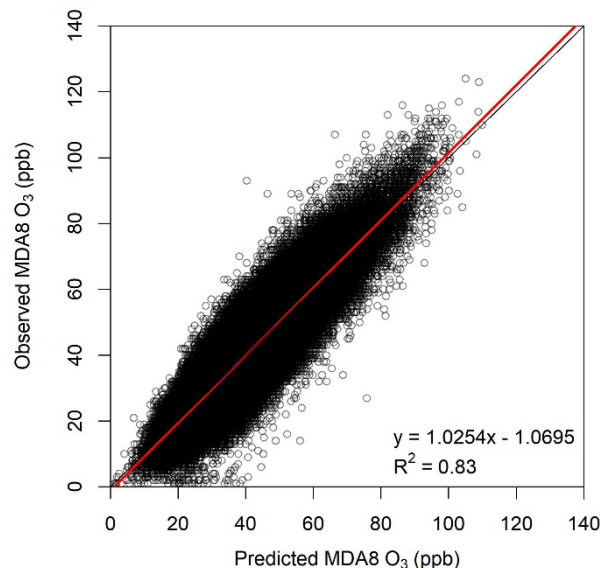


Figure 2. Observed vs predicted MDA8 for 803 sites, April–October 2019–2024. The residuals are defined by observed MDA8 – GAM predicted MDA8.

Figure 3 shows the distribution of GAM residuals for no smoke and smoke days. The mean residual is 0 (SD of 4.9 ppb) on non-smoke days and 3.8 ppb (SD of 7.8 ppb) on smoke days, which is indicative of the mean smoke contribution to the MDA8. We use the 97.5th percentile of the non-smoke residuals to evaluate whether smoke has made a significant contribution to the MDA8, based on published guidance from the US EPA (US EPA, 2016). In this case, the residual for an individual smoke day must be greater than the 97.5th percentile of the non-smoke residuals. If it is, then we can consider the MDA8 to have been significantly enhanced by smoke. The 97.5th percentile of the non-smoke residuals is calculated individually at each site. The overall mean 97.5th percentile at all sites is 9.8 ppb, with a range of 5.4–16.0 ppb. This procedure is shown in Figure 3.

Similar to the PM_{2.5} discussion above, one variable to pay attention to in our datasets is the “Quant_MDA8O3_resids”. This gives the percentile of each MDA8 residual, in comparison to the training (no smoke) data. The greater is this value, the greater is the confidence that smoke has influenced the O₃ MDA8.

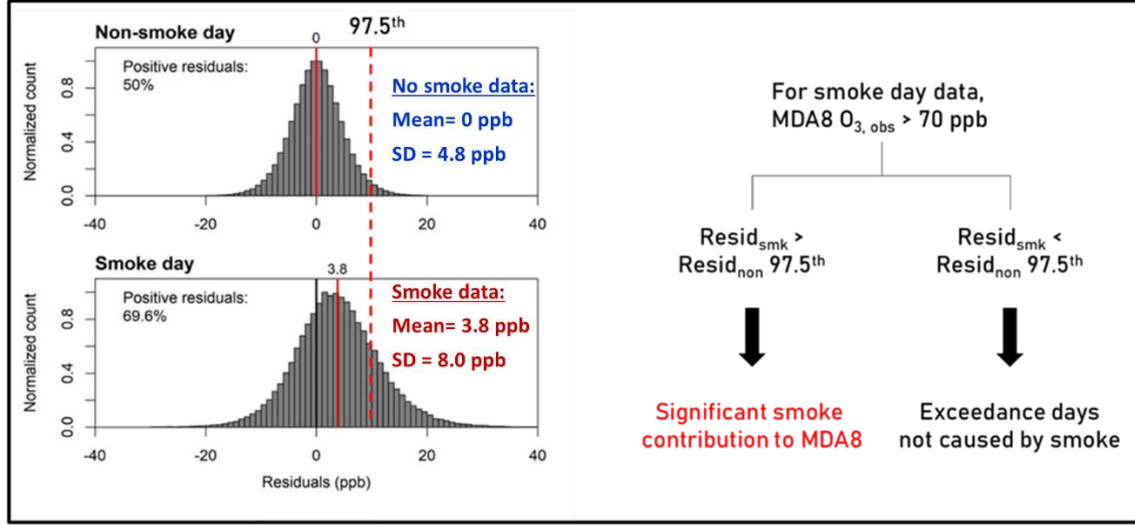


Figure 3. GAM residuals for no smoke and smoke days. The 97.5th percentile of the non-smoke day residual is used as a metric to evaluate smoke influence. We assume that any smoke residual that is greater than this 97.5th percentile is a day with a significant smoke contribution to the MDA8.

Empirical Distribution Matching (EDM) for bias correction in GAM:

Statistical and/or machine learning regression models are inherently biased because the model is fitted closer to the center of the response variable. This problem is usually not rigorously considered. However, bias correction can be a very important issue when selecting Exceptional Events Demonstrations (EEDs) by some criteria as shown in Figure 3. In particular, the 97.5th percentile of non-smoke days at each site can be changed when considering the bias correction, which may result in reassigning exceedance days with significant smoke contribution to MDA8.

Belitz and Stackerberg (2021) proposed an Empirical Distribution Matching (EDM) method to correct the bias of regression models. EDM corrects the bias by aligning the empirical cumulative distribution functions (ECDFs) of observed and predicted values as follows:

$$Y_{EDM} = Q_{Obs}^{Tr} \left(F_{pred}^{Tr} (Y_{pred}) \right)$$

Where “F” is the ECDF, “Q” is the inverse ECDF, and “Tr” represents the training data (i.e., non-smoke days data in this study). For each site, Q_{Obs}^{Tr} is obtained using observed MDA8, while F_{pred}^{Tr} is calculated using GAM predicted MDA8. Then, GAM predicted MDA8 (Y_{pred}) for both non-

smoke and smoke days is input into these functions, Q and F , to finally obtain Y_{EDM} . Hereafter, this method will be referred to as GAM-EDM.

Figure 4 shows performance of the GAM-v2-EDM for 803 sites nationally for data from April–October 2019–2024. The slope was slightly lower compared to Figure 2, and this is a natural phenomenon when applying bias correction. However, what is important is when we investigate true positives (TP) that exceed 70 ppb for both observed and predicted MDA8. The TP of GAM-EDM for training data (63%) improved by 18% compared to GAM (45%). As GAM-EDM adjusts predicted MDA8, the number of exceedance days with significant smoke contribution to MDA8 is slightly reduced. Being more conservative, GAM-EDM provides a more cautious estimate of exceedance days with significant smoke contribution to MDA8.

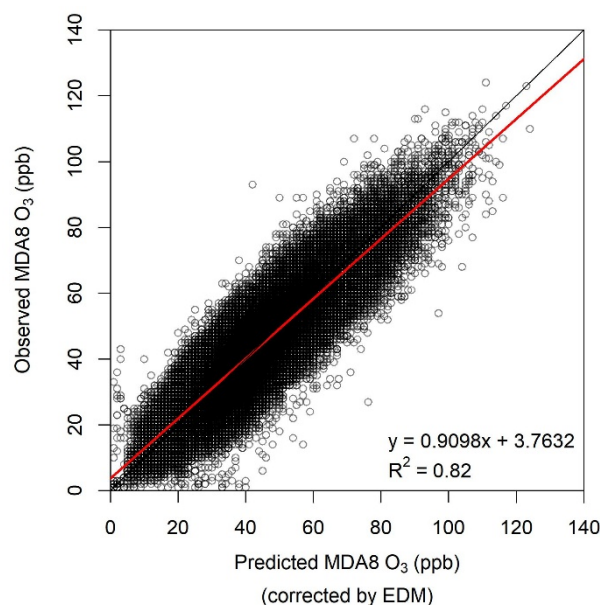


Figure 4. Observed vs predicted MDA8 for 803 sites, April–October 2019–2024. The figure also shows how we derive the residuals (observed MDA8 – GAM-EDM predicted MDA8).

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